

Functions and arguments

Project: Python readiness portfolio

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Lists and indexing

Without opening the old lesson, explain one key idea from Lists and indexing. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Branching and boundary tests

Without opening the old lesson, explain one key idea from Branching and boundary tests. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Values variables and types

Without opening the old lesson, explain one key idea from Values variables and types. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Follow the execution order for Functions and arguments. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Functions and arguments. Proposed explanation: “Writing def runs the body immediately”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Match positional arguments to parameters

Predict both returned and printed values, showing which argument supplies first and second. What is displayed if the program contains only the function definition and no calls? Why does swapping these arguments change the result?

```
def difference(first, second):  
    return first - second  
  
print(difference(11, 4))  
  
print(difference(4, 11))
```

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Label the parts

In `def show_remaining(capacity, used)`, identify the function name and parameters. In `show_remaining(15, 6)`, identify the arguments and their matching parameters.

My reasoning and evidence

6. W2 Follow the execution order

Using the correct definition, predict `show_remaining(8, 3)`, `show_remaining(8, 8)` and `show_remaining(8, 0)` in that order. Explain what happens if the file contains only the definition.

My reasoning and evidence

7. W3 Repair ignored arguments

The faulty body calculates `12 - 4` regardless of the call. State the output and expected output for `show_remaining(11, 2)`. Repair the expression and name an earlier call to rerun.

My reasoning and evidence

8. W4 Create a new function independently

Each sealed box contains 6 counters. Define `show_stock(boxes, loose)` to print the total counters using its two parameters. Assume nonnegative integer arguments. Call it with `(3, 2)`, `(0, 2)` and `(3, 0)`. Predict each output and explain why `show_stock(2, 3)` is a different case.

My reasoning and evidence

9. W5 State the input assumption

The correct subtraction function prints -1 for `show_remaining(4, 5)`. Does successful execution make this a valid cupboard record? Explain what the function assumes and what needs checking.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled functions and arguments evidence sheet to Python readiness portfolio. Show how you can distinguish a function definition from a function call. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Functions and arguments

Match arguments to parameters. A function prints the difference between capacity and reserved spaces.

Python 3: `def show_free(capacity,reserved): print(capacity-reserved)`. Calls: `show_free(11,3);`
`show_free(3,11)`.

Functions and arguments

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Define without calling

The definition creates the function but prints nothing yet.



2. Match first arguments

capacity=11 and reserved=3 give a printed 8.



3. Trace reversed call

capacity=3 and reserved=11 give -8 under the same arithmetic.

CHANGE ONE THING

Remove both calls and leave the definition only.

Explain the changed result using the matching lesson.

Match every argument to a parameter for the two calls. Why do outputs differ, and what prints when both calls are removed?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted outputs and the changed-case explanation. Use actual Python 3 results to check the prediction after reasoning first.

13. Project portfolio milestone

Project: Python readiness portfolio

Current milestone: Python readiness portfolio. Keep runnable code, predictions, actual outputs, one repaired error and an explanation of a function on changed inputs.

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted outputs and the changed-case explanation. Use actual Python 3 results to check the prediction after reasoning first.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Match arguments to parameters. A function prints the difference between capacity and reserved spaces.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? capacity=3 and reserved=11 give -8 under the same arithmetic.

1. Match first arguments
2. Trace reversed call
3. Define without calling

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Trace reversed call capacity=3 and reserved=11 give -8 under the same arithmetic. Positional arguments match parameter names in order. A function definition is not an execution of its body; separate calls supply separate input values.

Practice 2: explain the deciding evidence

Changed-case practice: Remove both calls and leave the definition only. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. We can decide the result without examining the changed input or the procedure.
3. No output is printed.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: No output is printed. The body runs only when the function is called.

Repair a missing building block

Review Lists and indexing (python-bridge/7). Goal: Distinguish a list index from the value stored at that position. Short practice: Label the three index positions. Predict all original outputs, then name the first output and stopping point after using index 3.. Return to this week and explain what you changed.

Review Branching and boundary tests (python-bridge/5). Goal: Trace the selected branch in an if elif else chain. Short practice: Trace -2, 4 and 5 through the branches. Explain why the negative check comes first and what label -2 gets if it is removed.. Return to this week and explain what you changed.

Review Values variables and types (python-bridge/2). Goal: Distinguish integer, floating-point, text and Boolean values. Short practice: Predict both print outputs and explain why they differ. Does count+tag work, and what happens to each valid output after count becomes ten?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
